PC26 · OBJECTIVE 1.1D

Shah Field De-Bottlenecking

Quantified opportunity catalogue — surface, subsurface & water injection

TotalEnergies ALSG × ADNOC Onshore

Asset Lead FP: Andrey Semenov

Issued 2026-04-23

CURRENT

75,733

BOPD actual

POTENTIAL

+10–12k

BOPD realistic 2026

OPPS

8

de-bottlenecking initiatives

OVERVIEW

Executive Summary

CURRENT FAR	2026 POTENTIAL	OPPORTUNITIES	DATA GAPS
75,733	**+10–12k**	**8**	**9**
BOPD actual	BOPD realistic	5 subsurface · 2 surface · 1 cross-cut	FCT report + IPM model top of list

KEY FINDINGS

SUBSURFACE

- ~32 strings flagged "Low PIP / stim required" by RM; 1.1C only covers 5
- 12 wells rate-capped to keep BHFP ≥ Pb+100 — drawdown headroom
- Inter-reservoir water loss confirmed on 139-SS & 145-SS
- R1 under-injected ~5 kbwpd vs sustainable VRR

SURFACE

- 22 strings ESP-limited not reservoir-limited (VSD-required)
- 14 strings already at 55–60 Hz — ESP upsize candidates
- Surface back-pressure: 10 psi separator reduction ≈ +800 BOPD
- FCT report pending — biggest tractable upside still unsized

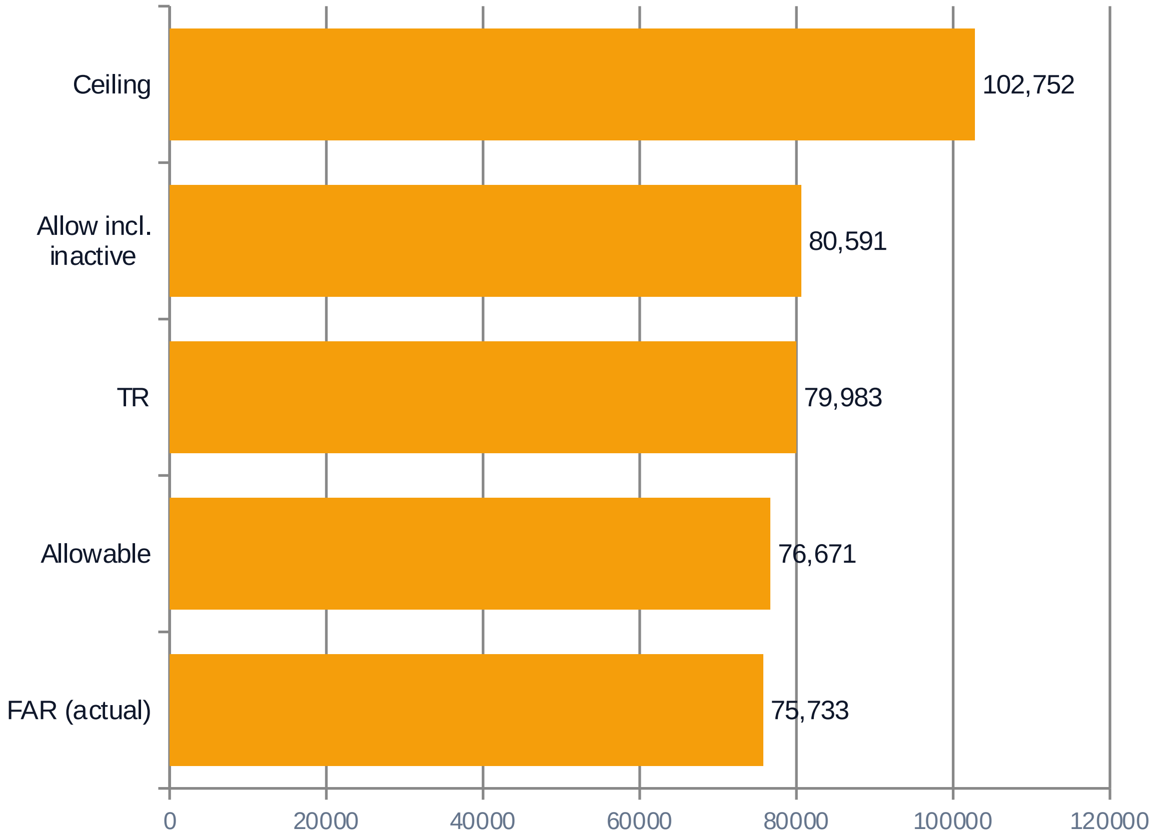
WATER INJECTION

- SY-175 & SY-180 stranded (16 kbwpd capacity) — WO in plan
- Currently 53 / 71 kbwpd vs technical rate
- Cluster #2 hard cap 25/26 Mbwpd binds Phase 2
- 12 high-WC producers cycle ~40 kbwpd of PW for ~960 BOPD

BASELINE

Field Baseline — Q2-2026 Allowable

OIL PRODUCTION (BOPD)



WATER INJECTION (bwpd)

Scenario	Cluster #1	Cluster #2	Total	Actual
With SY-175 + 180	26,000	24,600	**50,600**	—
Without SY-175/180	26,000	19,000	**45,000**	53,000
Technical rate	42,200	28,800	**71,000**	—
Cluster physical cap	26,000	25,000	*hard cap*	—

BASELINE READ Gap vs TR is 4,250 BOPD (5.6 %). Bigger prize sits between Allowable and the **deliverability ceiling of ~102,752 BOPD**, which represents the sum of estimated max rates for all active strings — and is unlocked only by capacity uplifts in subsurface and surface infrastructure. WI under-delivery (53 / 71 kbwpd vs technical rate) is the single biggest binding constraint at sector level.

CLASSIFICATION

Bottleneck Map — 9 root-cause categories

BOTTOM-UP FROM RM COMMENT COLUMN

SUBSURFACE

S1	Low PIP / stim candidate	32 wells	12,000 BOPD
S2	High water cut (>60 %)	38 wells	7,000 BOPD
S3	Low BHFP — Pb risk	12 wells	1,800 BOPD
S4	Inactive — WO backlog	14 wells	3,920 BOPD
S5	Reservoir P depletion	8 wells	5,000 BOPD

SURFACE / COMPLETION

C1	Low DP / VSD required	22 wells	6,500 BOPD
C2	ESP at max frequency	14 wells	3,500 BOPD
C3	Back-pressure (flowline)	TBD	~2–4 kBOPD est.

WATER INJECTION

W1	Cluster cap 26 Mbwpd	–	physical ceiling
W2	WI integrity 175/180	2 wells	16 kbwpd stranded
W3	R1↔R2 water loss	2 wells	tracer needed
W4	High-WC producers	12 wells	PW loading

Note: Counts and BOPD-lost figures are RM-classified analytical estimates (§ANALYSIS BLOCK). C3 surface back-pressure pending Jan-2026 FCT report.

PRIORITIZATION

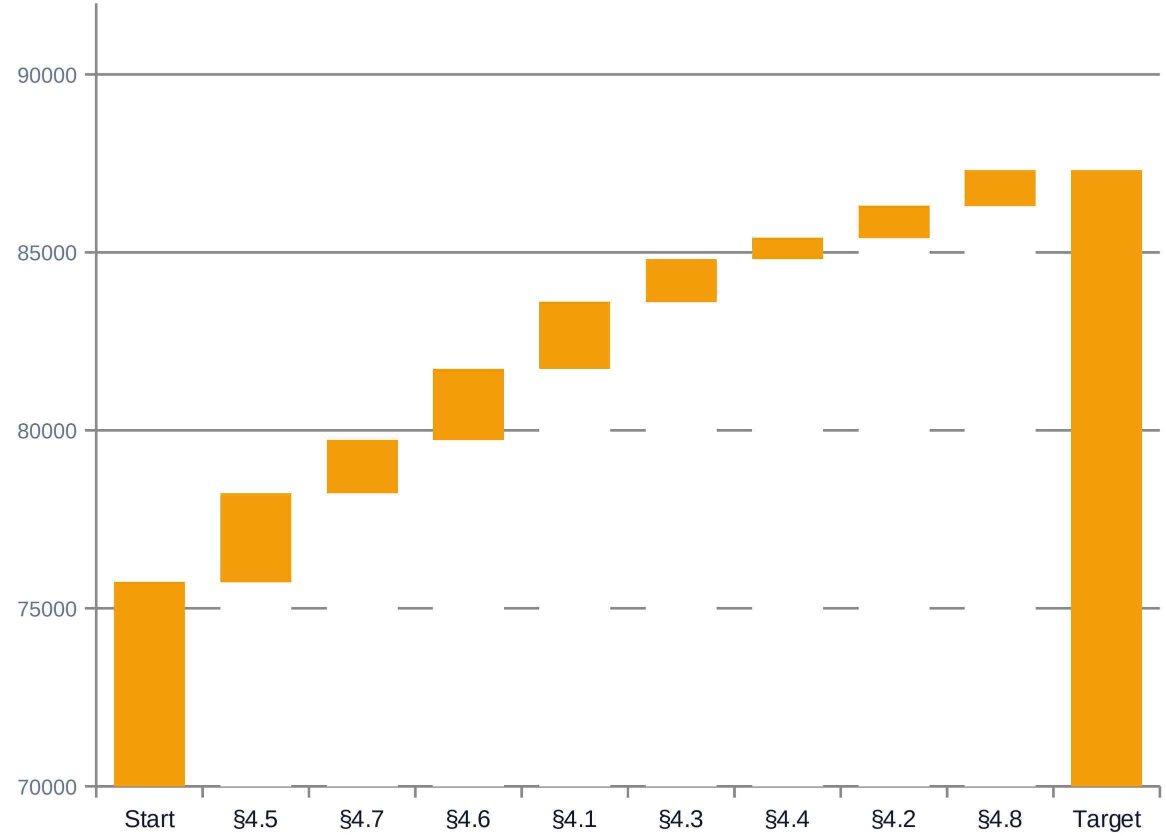
Consolidated Ranking & Gain Bridge

RANKED BY 2026 REALISTIC GAIN

#	ID	Opportunity	2026 BOPD	Conf.	Capex
1	§4.5	WI capacity (SY-175/180 + cluster)	2,500–4,000	M-H	$2–28M
2	§4.7	Surface back-pressure optimization	1,500–2,500	M	$0.2–8M
3	§4.6	PW redistribution + R1/R2 rebalance	2,000–2,400	M	$0.5–1.5M
4	§4.1	Stimulation roll-out beyond 1.1C	2,500	M	$8–12M
5	§4.3	VSD / PWS retrofit campaign	1,600	H	$2.4–4.8M
6	§4.4	ESP upsize (piggyback WOs)	800	M	$2.0–3.2M
7	§4.2	BHFP rate-cap relaxation	900	M-H	$0.2–0.5M
8	§4.8	Inactive-string reactivation	1,000	H	$0.5–1.0M

TOTAL realistic envelope (×0.75 overlap deduction) ~9,600 BOPD

GAIN BRIDGE — 75,733 → ~87,000 BOPD



+11,575 BOPD cumulative — within 10–12 kBOPD realistic envelope.
§4.1+§4.3+§4.4 deflated ×0.75 for overlap.

Saih Rawl Shuaiba analog (SPE 68222): +6,300 BOPD on smaller system via WI rebalancing + automation — directly supports the §4.5/§4.6/§4.7 envelope.

OPPORTUNITY DEEP-DIVE

§4.5 — WI capacity uplift (SY-175 + SY-180 + cluster review)

2026 REALISTIC GAIN

2,500–4,000

BOPD

FULL POTENTIAL

4,000–5,500

BOPD

CAPEX BRACKET

$2–28M

$M · Med-High

MECHANISM

Phase 1 (Q2/Q3-2026): Complete WO on SY-175 & SY-180 to recover 16,000 bwpd stranded capacity. RM-mandated prerequisite: FCT in Cluster #2 / SY-148 WS before SY-180 commissioning to confirm Cluster #2 can handle the additional load (currently 19/28.8 kbwpd).

Phase 2 (2027): Cluster #2 capacity uplift study — if FCT confirms hard ceiling at 26 Mbwpd, evaluate pump/manifold capacity to lift to 35 Mbwpd.

Phase 3 (long-term): New WI well drilling if VRR confirms additional voidage to fill. Saih Rawl analog (SPE 68222): ~12 % oil rate uplift per 30 % WI rate increase.

PREREQUISITE

Cluster #2 / SY-148 WS FCT (single most important data request) · G2V2 ML#5 reservoir model · Voidage Replacement Ratio analysis

RECOMMENDATION

Approve Phase 1 SY-175/180 WO execution immediately (capex already in plan). Commission FCT in CL02/SY-148 as RM-mandated prerequisite before SY-180 start-up. Stage Phase 2 cluster uplift decision pending FCT results.

OPPORTUNITY DEEP-DIVE

§4.7 — Surface back-pressure optimization

2026 REALISTIC GAIN

1,500–2,500

BOPD

FULL POTENTIAL

2,300–3,300

BOPD

CAPEX BRACKET

$0.2–8M

$M · Medium

MECHANISM

Phase 1: Station header pressure setpoint review at CDS / RDS-1 / RDS-2. Each 10 psi separator reduction across 80 R1 producers ≈ +800 BOPD via reduced WHFP and improved IPR/VLP intersection.

Phase 2: Selective flowline looping on 2-3 worst segments (>200 psi/km ΔP). Phase 3: Choke-on-test vs production-choke alignment — many wells tested at higher chokes than routinely operated.

Direct evidence pending FCT, but proxy indicators (full choke + ESP at 50 Hz + rate plateau, WHFT 150-162 °F) point to downstream restriction.

PREREQUISITE

January-2026 FCT report (single most important data request after Cluster #2 FCT) · GAP/Prosper IPM model · WHIP/WHFP-vs-time per well

RECOMMENDATION

Hold quantification at pending-FCT estimate. Once FCT report lands, commission GAP/Prosper rerun with revised separator setpoints. Phase 1 is near-zero capex — pursue setpoint review as fast win.

OPPORTUNITY DEEP-DIVE

§4.6 — Produced-water redistribution + R1↔R2 rebalancing

2026 REALISTIC GAIN

2,000–2,400

BOPD

FULL POTENTIAL

2,400–3,200

BOPD

CAPEX BRACKET

$0.5–1.5M

$M · Medium

MECHANISM

(a) Inter-reservoir tracer + integrity diagnosis on 139-SS and 145-SS — RM explicitly flagged R1→R2 water loss; if confirmed, cement squeeze to restore zonal isolation.

(b) Rebalance injection R1/R2: R1 currently under-injected ~5 kbwpd vs sustainable VRR. Redirect 3-5 kbwpd from R2 to R1 once Cluster #2 integrity allows.

(c) Selective shut-in of 6 worst high-WC producers (>85% WC) — collectively producing ~960 BOPD oil but consuming 40 kbwpd PW handling capacity. Net benefit: +1,200 BOPD elsewhere via better VRR.

PREREQUISITE

PW lab analysis (TDS, scaling, particulates, H_2S) · Injector PLT / CBL on 139-SS, 145-SS · G2V2 rerun with rebalanced injection

RECOMMENDATION

Greenlight low-capex tracer campaign immediately ($0.5-1.5M). Concurrent: identify the 6 high-WC shut-in candidates from the Wells table for FY-2026 decision. PW lab work is independent and can start at any time.

OPPORTUNITY DEEP-DIVE

§4.1 — Stimulation roll-out beyond Obj 1.1C

2026 REALISTIC GAIN

2,500

BOPD

FULL POTENTIAL

5,000–6,000

BOPD

CAPEX BRACKET

$8–12M

$M · Medium

MECHANISM

32+ R1 strings flagged "Low intake / Low PIP / stim required" by RM. Obj 1.1C delivers stimulation on only 5 wells in 2026.

Apply proven Naphtha+HCl recipe to 8-10 additional R1 wells per year as sustained program. Target wells with PI degradation D > 30 %/yr (1.1C methodology). Average gain +150 to +400 BOPD per well, declining 30 %/yr post-job.

Full backlog of 25 incremental candidates × 250 BOPD avg ≈ +6,000 BOPD at peak if all done within 18 months. Realistic 2026 deliverable: 10 incremental wells = +2,500 BOPD.

PREREQUISITE

Per-well PI screening across 32 stim candidates (extension of 1.1C dashboard methodology)

RECOMMENDATION

Extend the Obj 1.1C dashboard with PI-degradation screening across all 32 RM-flagged candidates. Build Q3-2026 stim queue from top-D ranking. Schedule 10 wells for 2026, decision-gate the remaining 15 against 2026 results.

EXECUTION PLAN

Data Gaps & Next Steps

CRITICAL DATA GAPS

01 **January-2026 FCT report**
§4.7 cannot move past estimate

02 **GAP/Prosper IPM model**
Back-pressure sensitivities

03 **G2V2 reservoir model — ML#5**
§4.5 oil response · §4.6 rebalancing

04 **Cluster #2 / SY-148 WS FCT**
RM-mandated for SY-180 commissioning

05 **PIDs / PFDs / vendor sheets**
CDS, RDS-1/2, WI clusters

06 **Produced water lab analysis**
§4.6 chemical design

07 **WHIP/WHFP-vs-time per well**
§4.7 flowline hypothesis

08 **Injector PLT/CBL (139-SS, 145-SS)**
§4.6 (a) integrity diagnosis

09 **Per-well PI screening (32 candidates)**
§4.1 backlog scoping

EXECUTION TIMELINE

Q2-26
- SY-175/180 WO start
- PW lab campaign
- FCT submission

Q3-26
- Tracer 139/145-SS
- VSD retrofit kick-off
- Tech WS sign-off

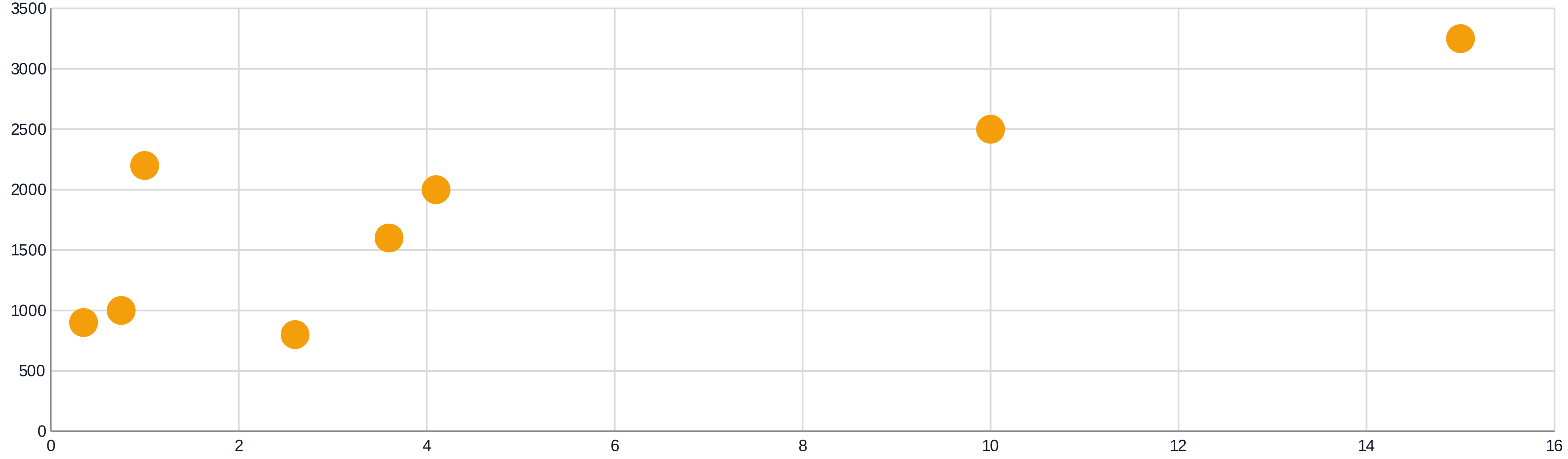
Q4-26
- SY-180 commissioning
- Setpoint pilot
- Stim batch 1 (5 wells)

2027
- Cluster #2 uplift
- Stim batch 2 (5 wells)
- ESP upsizes

Capex vs Gain — Prioritization Matrix

TOTAL CAPEX RANGE	2026 GAIN MIDPOINT	$ / BOPD AVERAGE	PAYBACK @ $80/bbl
$16–58M	**~11,575**	**$3,200**	**< 6 mo**
across 8 opportunities	BOPD with overlap	weighted across all opps	on top 4 opportunities

CAPEX (X) vs GAIN (Y) — bubble = $/BOPD



§4.6 (PW redistribution) and §4.2 (BHFP relaxation) sit in the sweet spot: low capex, fast gain. §4.5 has the widest capex range because Phase 2 cluster uplift is contingent on FCT.

CONCLUSIONS

Shah field can lift 10–12 kBOPD within 12 months

with the right sequencing of subsurface, surface, and WI initiatives

01 **BIGGEST ABSOLUTE PRIZE IS WATER INJECTION**

§4.5 SY-175/180 + cluster review delivers 2.5–4 kBOPD realistic — the largest single contribution and a pre-requisite for sustained R1 pressure support. Already partially budgeted via WO plan.

02 **SURFACE BACK-PRESSURE IS THE NEXT-LARGEST AND CHEAPEST WIN**

§4.7 phase-1 setpoint review costs ~$0.2M and could deliver 1.5–2.5 kBOPD. But the entire envelope is hostage to the Jan-2026 FCT report and GAP/Prosper IPM model — both still missing. Single biggest data request.

03 **PRODUCED-WATER REDISTRIBUTION IS A NEAR-ZERO-CAPEX OPP**

§4.6 R1↔R2 rebalancing + tracer integrity diagnosis on 139-SS/145-SS delivers ~2 kBOPD at $0.5–1.5M. Lowest-friction commercial recommendation — can start before FCT report lands.

04 **STIMULATION BACKLOG IS DEEPER THAN OBJ 1.1C SCOPE**

~32 R1 strings flagged "Low PIP / stim required" by RM. Obj 1.1C addresses 5. Sustained program of 8-10 wells/year via Naphtha+HCl recipe builds a +6 kBOPD asset over 18 months at $8-12M capex.

05 **NINE CRITICAL DATA GAPS — FCT REPORT IS #1**

Until the Jan-2026 FCT report, the GAP/Prosper IPM model, and the G2V2 ML#5 forecast are received, §4.7 stays at "pending-FCT estimate" and §4.5 Phase 2 cluster uplift decision cannot be quantified.